
VOCODER BUILDING GUIDE

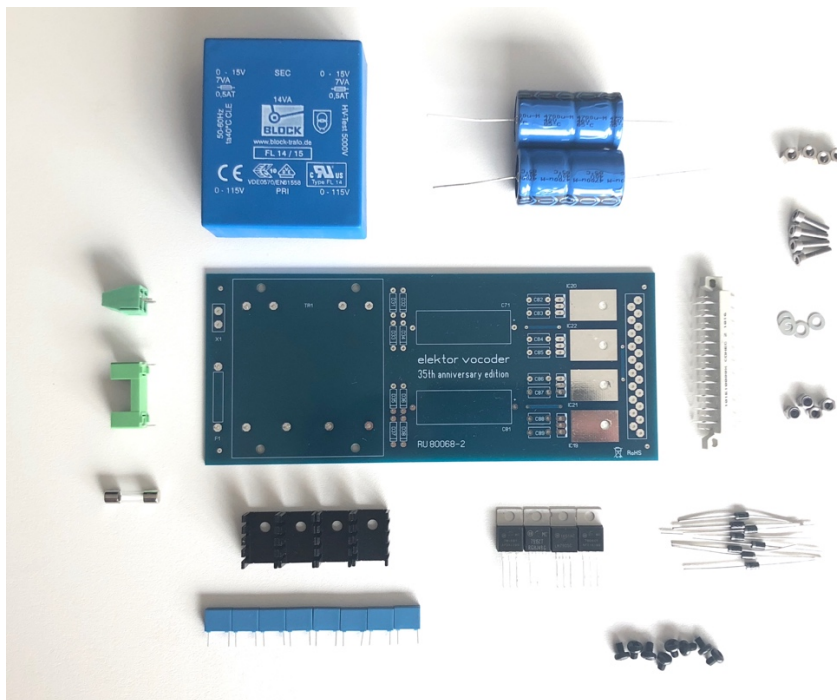
Edition 1.2

Step 1A

Power Supply - RU80068.2

Estimated building time: 1,5 hour

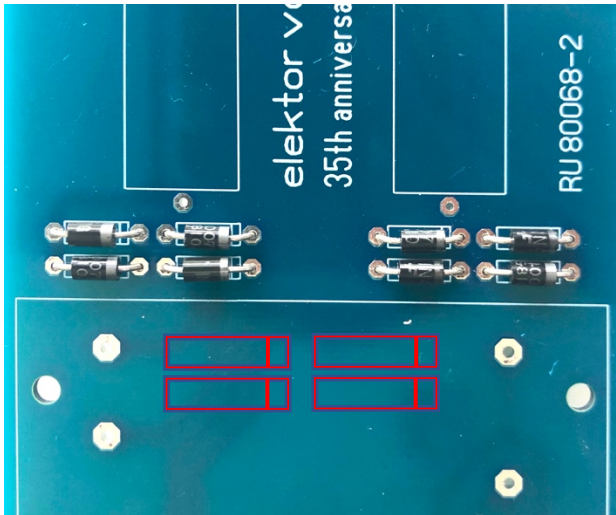
- Sort all required components as indicated in the BOM:



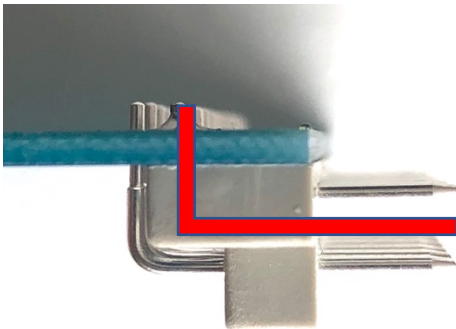
- Bend the diodes so that they fit neatly right away:



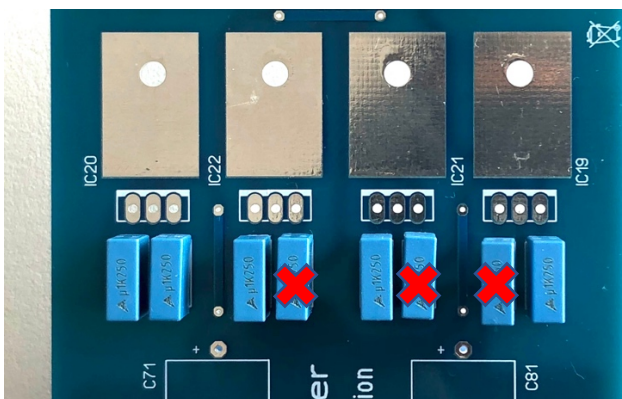
- Solder the 8 diodes, note the direction of the white line (component) and tabbed indication (silkscreen). Do not reverse the polarity of these components:



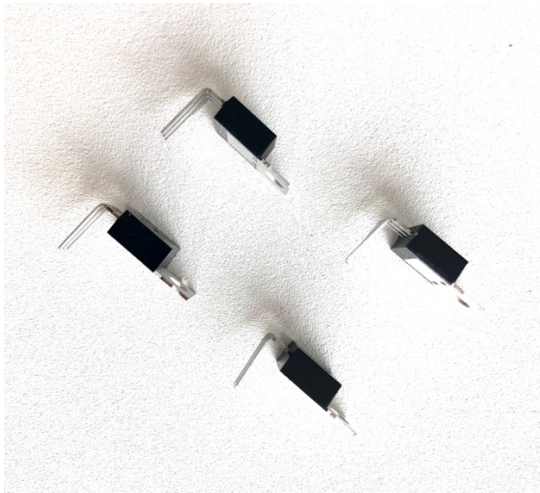
- Put the 21 pin connector to the board, do not exert any excessive force. Solder only pin 1 and pin 21 and check for a right angle. Correct if necessary and solder all remaining pins:



- Place the 100 nF capacitors and start soldering only 1 of both legs. This allows lining up the components. Correct if necessary and solder all remaining legs:



- Bend the power regulators so that they fit neatly right away:



- Fasten the PCB, heat sinks and regulators together with nuts & bolts and do not forget to apply some washers. Thermal paste is not necessary due to the limited dissipation in the regulators. Align the heat sinks before tightening the nuts & bolts.

Check twice the right regulator is on the right place:

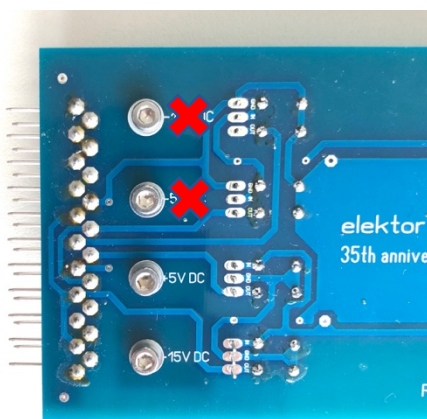
IC19 = 7815

IC20 = 7905

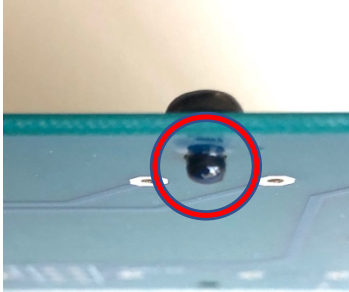
IC21 = 7805

IC22 = 7915

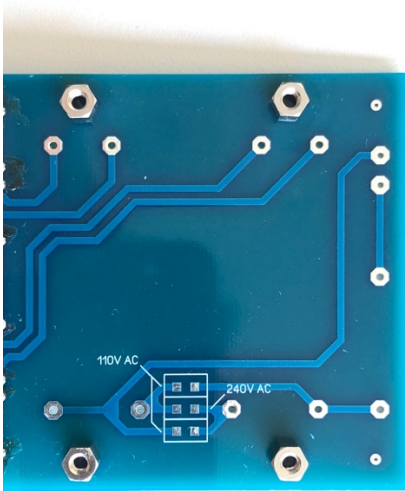
On some boards the -15V DC & -5V DC indication on the solderside is exchanged, so do not use this indication as reference. Only rely on the above regulator list to put the components in place:



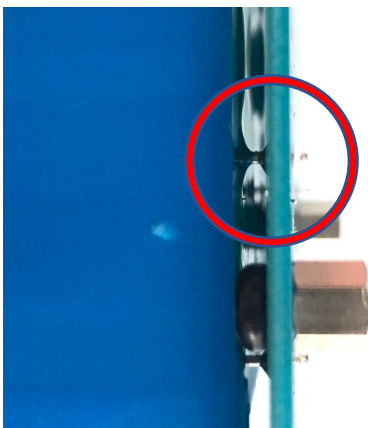
- Apply some Loctite on the bolts to fix the power supply cover plate. The hex bushing is not longer accessible after soldering the transformer, so do not skip this step. Superglue can do the job as well, but loosening becomes very difficult:



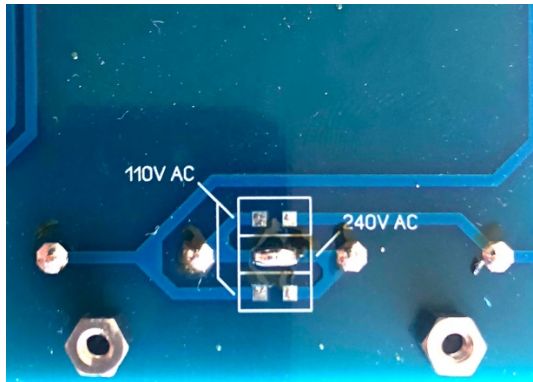
- Fasten the 4 screws and bushings:



- Solder the transformer on the PCB, be careful while bending the pins as they will break down if you are too rough. Apply pretty much solder and let it flow through the plated hole as this will fix the transformer also on the component side of the PCB:



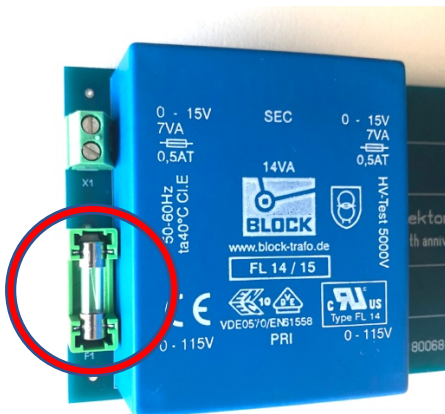
- Select the mains voltage of your country by bridging the jumpers on the right position. For 110V AC you 2 bridges should be applied, for 240V AC only one bridge should be closed. Check twice before you proceed:



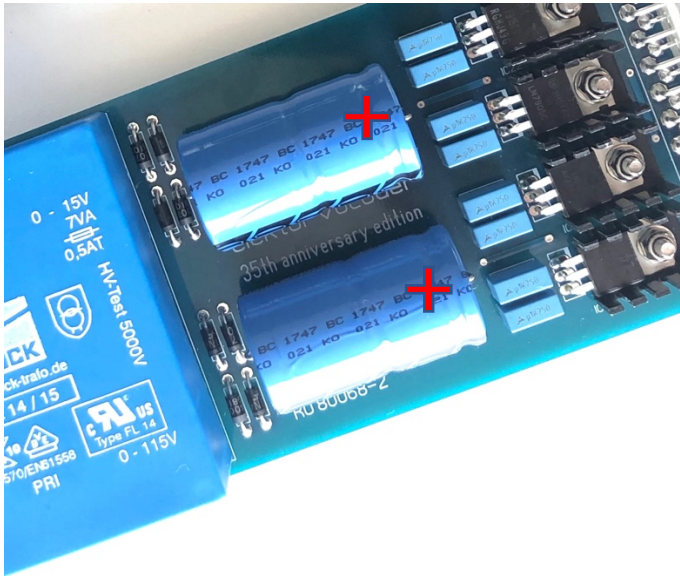
- Insert the AC terminal block and make a solid solder connection. Depending on the chosen model it may happen that the connector legs are square, in that case use a pliers to squeeze the pins a little. Do not exert any excessive force to avoid damaging the plated holes:



- Put also the fuseholder and fuse in place. Do not forget to cover this, otherwise mains voltage can be touched:



- Bend the electrolytic capacitors (e-caps) and put them in the right position. Do not reverse the polarity of these components:



- Thoroughly clean the PCB using flux cleaner, try not to touch the the transformer as the white imprint will disappear.

Step 1B

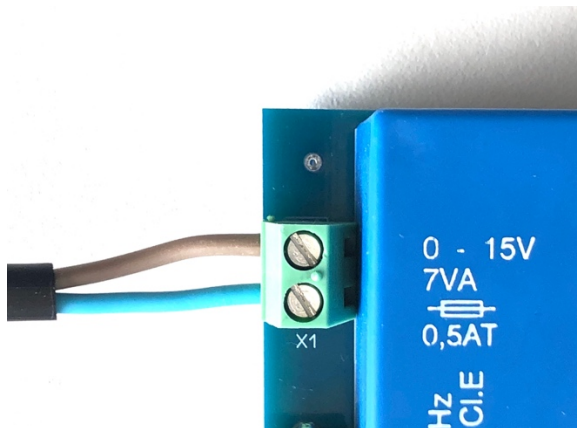
Power Supply Cover - RU80068.0

Estimated building time: 10 minutes

- Place the power supply cover over the AC current parts and screw the lid in place.



- Check all above steps and temporary connect a power lead to the AC terminal block:



The tension over both electrolytic capacitors (e-caps) should be around 25V DC. While measuring this tension check also the polarity of the capacitors. Immediately power off the device if there is a mispolarisation.

- Your power supply is now ready:

